

Recycle or Reuse

There are other alternatives, besides recycling computers. For instance, charitable organisations collect them and after they are inspected, consider them for reuse. It would be a waste of resources if you recycle a working computer. Though slower speeds of older machines might be considered a hindrance for commercial or scientific organisations, they can be used in schools and other institutes for educational purposes where speed is not the main criterion.

There are several organisations working with this idea. In India, Mumbai-based FORCE (Foundation for Computer Education) collects surplus or obsolete computers from corporates. After inspecting them and repairing them if necessary, it gives them to schools and colleges registered with the organisation. Notably, a second-hand computer is not a rare thing in India. Computers are seldom discarded in India, unless of course it has reached a point of no return. In fact, many hardware shops in major cities carry out a thriving business refurbishing old computers.

Most states in the US have at least one reuse programme, which caters to the need of local schools for computers. These organisations collect obsolete equipment from big business organisations, refurbish them and donate them to the needy schools.

Students Recycling Used Technology (StRUT) is a specialised programme in which the participating schools' students evaluate donated computers and decide which components can be reused and which recycled.

industry estimates, PC users trash approximately 4 million floppy disks everyday. This magnetic material takes between 450 to 500 years to get degraded. Hence, companies all over the world are working towards greener initiatives for recycling floppy disks.

GreenDisk, founded on World Earth Day takes all discarded disks to their plant in Columbia, where the media is fully inspected and magnetically erased. The disks are then disassembled and the metal and plastic components are recycled to make new disks and other items. To date, GreenDisk has recycled more than 20 million disks. Thus, software companies are also benefitted—they don't have to pay for dumping the outdated packs and are relieved of the storage costs.

Old goods for new

Computer manufacturers are introducing schemes wherein users can exchange their old computers for new ones. In turn, the manufacturers recycle the PCs that they receive from their customers. IBM, Intel, Sony and HP are leaders in this arena.

Recently, HP announced that it would accept used equipment of any manufacturer for recycling. HP recycles almost all components such as inkjet supplies, laser jet supplies, and other hardware that it receives from customers and from within HP. Till date, HP has recycled around 39 million laser jet cartridges and it has reduced the use of plastics in the production of cartridges by 44 per cent, allowing them to be easily recycled when trashed.

Products that are produced from recycled plastics include buttons, plastic trays, tractor feed parts and other items used in the production of printers and its components.

HP Inkjet products are also recycled to produce automobile parts, trays used in microchip fabrication processes, copper wire, steel plates and other precious metals used in electrical and electronic equipment. Other parts, which cannot be recycled, are disposed of in an environmentally safe manner. The company has a separate section on its Web site for selling refurbished products, which are priced lower than ones that are not.

Sony, another major player in the electronics industry, has also joined the green movement. It is operating in conjunction with the state of Minnesota (US) to find the best strategies to dispose of electronic waste safely. The plan is to collect material, find economically feasible methods to



Scientist at PRL working on recycling

recover value from them, and use these strategies for drawing inferences for future operations.

IBM, the big blue, has also offered its individual consumers and businesses a means to recycle their outdated hardware. Using this service, you can recycle any brand of PC (and this includes monitors, printers and other peripherals) for an economically feasible price of \$29.99 (approx Rs 1,300), including shipping.

Envirocycle is the name of the dedicated recycling unit based in the US. The company tries to make the maximum possible use of a PC and its components. Repaired PCs are donated to Gift for Kind International (GIKI), a charitable organisation that contributes to over 50,000 non-profit organisations all over the world. The PC recycling service can be purchased from IBM when buying a new IBM PC as a standalone service.

Is the future green?

Reuse? Yes. Recycle? Yes. If not either, then what? One answer is to dispose things in an environmentally safe way. Besides, we can think of ways to reduce the utilisation of toxic components in the making of a computer. Redesigning the PC could be the best alternative that would aid recycling in the future.

However, there are some obstacles that make this task slightly difficult. For instance, lead is used in monitors to prevent radiation and hence cannot be drastically reduced. Its use in the front panel of monitors has been virtually eliminated, but the industry has been less successful in doing away with the metal in other parts of the monitor.

Though the introduction of flat panel monitors would gradually reduce the use of CRTs (which have lead), it would result in the introduction of the equally toxic metal, mercury. IBM has reduced the amount of screws, bolts, plastics and glues in the making of its products, which will make it easier for recyclers to disassemble and process old computers.

But there will still exist an element of pollution, which cannot be fully removed as long as computing technology plays a vital role in our lives. It might be next to impossible to make all computer components pollution free. We'd have to either reduce the usage of computing technology or find ways to dispose them in an environmentally safe manner. Since the former is not possible in the digital age, we must recourse to the latter. ■